

DATA3: Single-Salt NF270 Diafiltration Analysis

Collaborator update • June 24, 2026

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Feedback from Friday — addressed

- **Time-series correction that matches DATA2 conventions:**
Mass, retentate, and permeate share one time axis
- **Identifiability contours in parameter space:**
 σ vs. B log-WSSE contour surface plots generated per experimental run - new plots being generated for DATA3 -
 - Mass,
 - ICP-based permeate + retentate concentration,
 - Conductivity probe-based permeate + retentate data, and
 - Conductivity probe-based permeate + retentate concentration) at the identified L_p
- **Color coding:** fit/data plots follow the DATA2 convention (retentate vs permeate vs model) across all the plots

Topics I need help with

Prioritization based on impact:

1. Accuracy of process model: Extended Nernst-Planck Equation

$$J_i = v_s \cdot C_i \cdot K_{(i,v)} - D_i \cdot \nabla C_i - z_i \cdot C_i \cdot D_i \cdot F / RT \cdot \nabla \psi$$

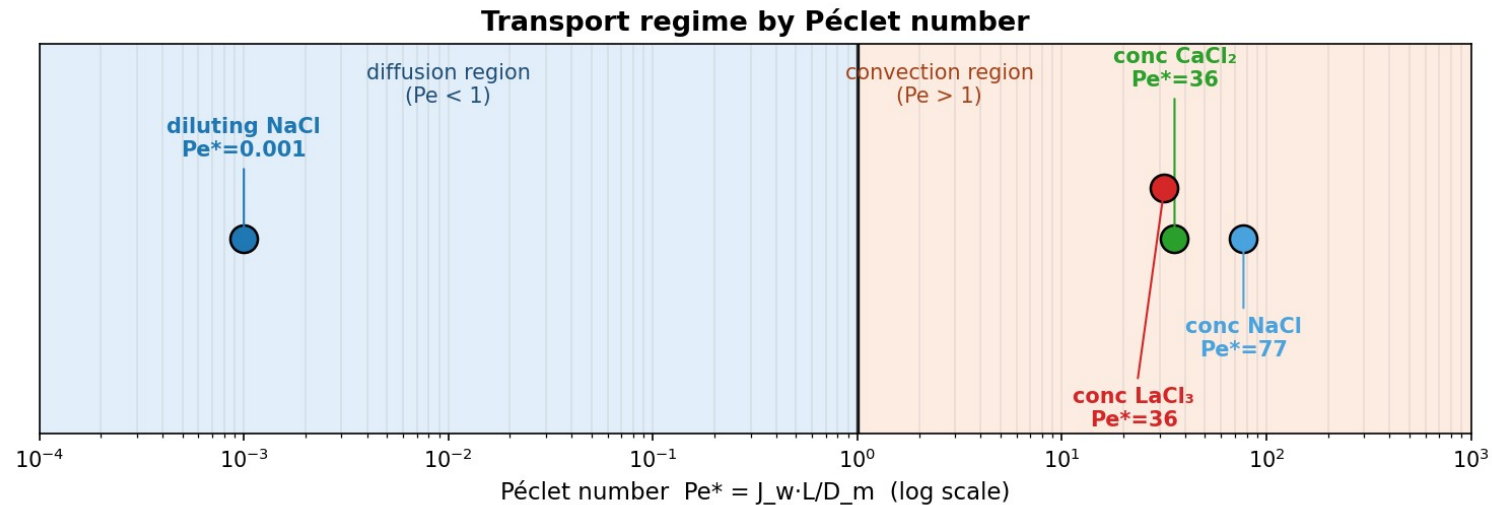
2. Interfacial concentration (supported by plots): What range of membrane-interface concentration do the experiments actually probe? How does this impact our correlation B with the interfacial concentration?

3. Peclet framing (supported by plots): The Peclet number and the rejection law $R(\sigma, Pe)$ - Is this the right perspective?

4. Transport components yet to be modeled: cross-coefficients D_{ij} , $B(c)$, and activity coefficients — which to include?

- **Péclet framing:** the Péclet number and the rejection law $R(\sigma, Pe)$ — is this the right lens?
- **Transport components:** dielectric exclusion, cross-coefficients D_{ij} , $B(c)$, and activity coefficients — which to include?
- **Identifiability:** given σ is unidentified single-experiment, how far can $B(c)$ claims go?

The same membrane spans diffusion- and convection-limited regimes



$$Pe = \frac{\text{convection}}{\text{diffusion}} = \frac{J_w L}{D_m}$$

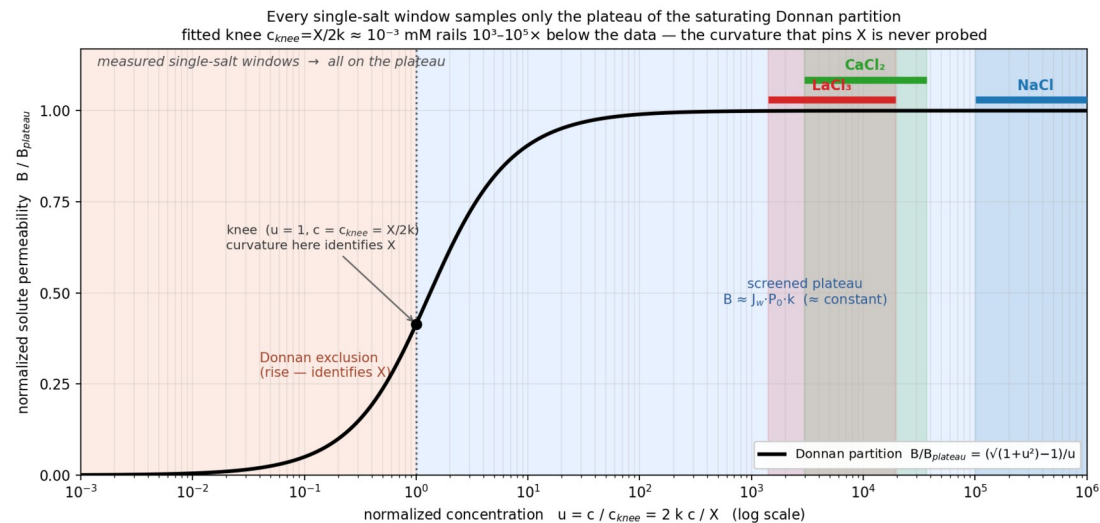
$$R = \frac{\sigma(1 - e^{-Pe})}{1 - \sigma e^{-Pe}}$$

Plotted Pe^* = the regression-optimal Péclet per experiment (argmin over Pe of the convection-diffusion partition-fit MSE) — a fitted regime indicator, NOT $J_w \cdot L / D_m$ evaluated directly. **where** Pe — Péclet number; J_w — water flux; L — membrane thickness; D_m — solute diffusivity; R — salt rejection; σ — reflection coefficient (0–1).

- **Diluting NaCl is diffusion-limited ($Pe^* \approx 10^{-3}$); concentrating NaCl / CaCl_2 / LaCl_3 are convection-dominated ($Pe^* \approx 77 / 36 / 36$).**
- **Mechanistic inference, not a direct demonstration:** Pe and B come from the same fitted model — this motivates $B(c)$, it does not prove it.
- **To tabulate for review:** water flux J_w , membrane length-scale L , and diffusivity D_m per salt.

Refs — $Pe = J_w L / D_m$: DATA2, Liu et al., *Ind. Eng. Chem. Res.* 64 (2025) 12111 (eq 25); Spiegler–Kedem rejection: Spiegler & Kedem, *Desalination* 1 (1966) 311; solution-diffusion (L_p , B , σ): Kedem & Katchalsky, *Biochim. Biophys. Acta* 27 (1958) 229

All single-salt windows sit on the Donnan plateau



$$B(c) = \frac{J_w P_0 (-X + \sqrt{X^2 + 4k^2 c^2})}{2k}$$

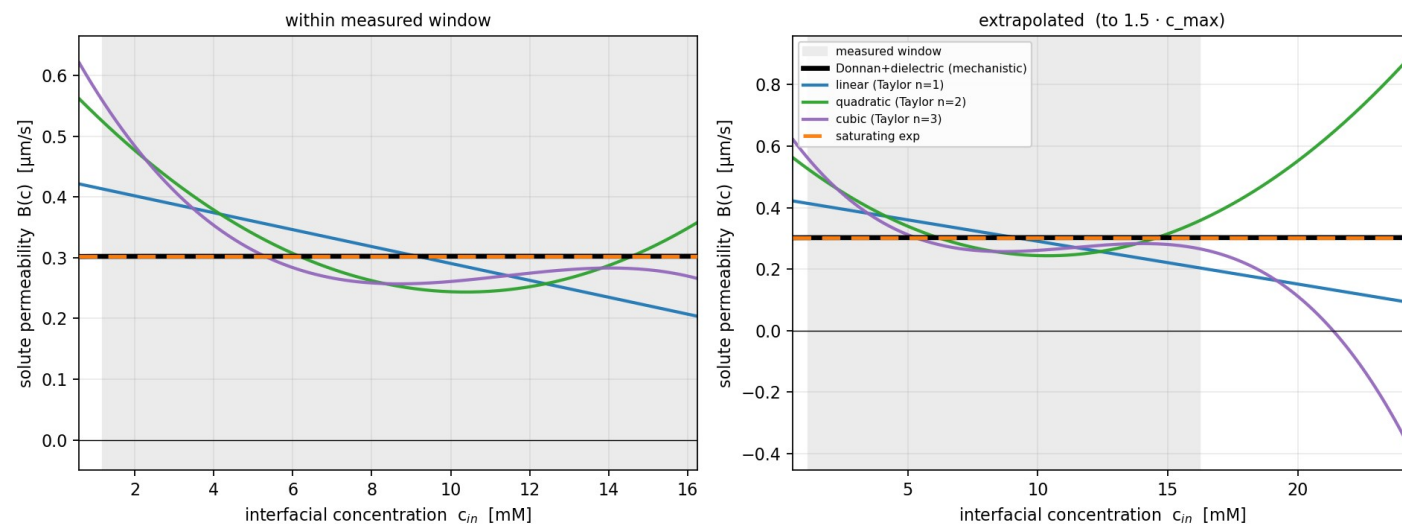
$$B_{plateau} = J_w P_0 k, \quad c_{knee} = \frac{X}{2k}, \quad u = \frac{c}{c_{knee}} = \frac{2kc}{X}$$

where $B(c)$ — solute permeability at interfacial concentration c ; c — membrane-interface salt concentration; J_w — water flux; P_0 — intrinsic permeability prefactor; X — effective membrane fixed-charge density; k — lumped Donnan-dielectric partition factor; $B_{plateau}$ — high- c plateau value; $c_{knee} = X/2k$ — regime knee; $u = c/c_{knee}$ — screening parameter.

- **$B(c)$ collapses onto one master curve; every measured window is at $u \approx 10^3$ - 10^5 (screened plateau) → B is $\approx$ constant in-window (the screened-plateau value).**
- **The knee $c_{knee} = X / 2k$ rails 10^3 - $10^5 \times$ below the data → the fixed-charge scale X is not identified from one salt.**

Refs — Donnan-dielectric $B(c)$: Yaroshchuk, Adv. Colloid Interface Sci. 85 (2000) 193; Bandini & Vezzani, Chem. Eng. Sci. 58 (2003) 3303 (DSPM-DE); Donnan-equilibrium partition cf. Bolt, J. Colloid Sci. 10 (1955) 206 (as in DATA2).

B(c): in-window agreement, out-of-window divergence — LaCl_3



$$B(c) = \beta_0 + \beta_1 c + \beta_2 c^2 + \beta_3 c^3$$

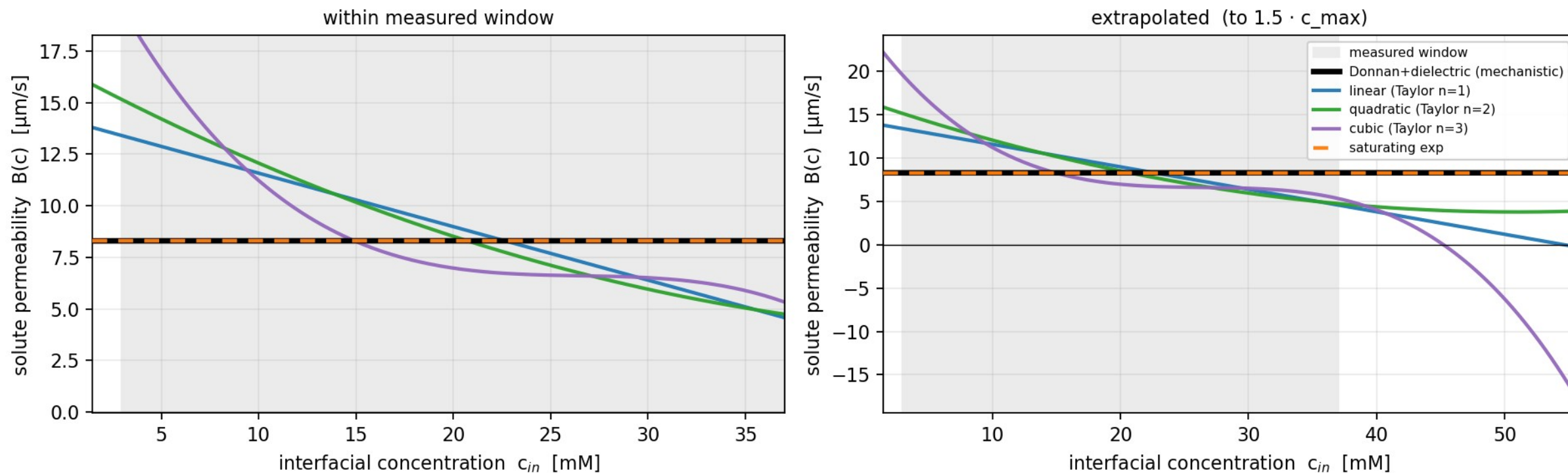
$$B(c) = B_\infty (1 - e^{-c/c^*})$$

where $B(c)$ — solute permeability; c — interfacial concentration; β_0 – β_3 — polynomial (Taylor) expansion coefficients; B_∞ — saturating permeability as $c \rightarrow \infty$; c^* — characteristic saturation concentration.

- Within the measured window, constant / linear / quadratic / cubic / saturating-exp all agree ($B \approx 0.3 \mu\text{m}\cdot\text{s}^{-1}$).
- Extrapolated, the polynomials diverge unphysically (cubic $\rightarrow$ negative); Donnan and saturating-exp stay bounded.
- The Donnan-dielectric is the mechanistically-grounded, physically-bounded form — the data do not discriminate it from a constant in-window.

Refs — empirical $B(c)$: polynomial (multiplicative) form per DATA2, Liu et al., Ind. Eng. Chem. Res. 64 (2025) 12111 (eq 31); saturating-exp & Donnan-bounded forms: this work (DATA3).

Same behavior for CaCl_2



- **Identical picture: in-window agreement, out-of-window polynomial divergence, Donnan bounded.**
- **On parsimony (AIC) a constant fits as well in-window** — prefer Donnan for mechanism and physical extrapolation, not in-window fit.
- **Next step to pin X:** a joint multi-salt fit (re-parameterized by ionic strength) reaches lower u and probes the knee.

Backup Slides

Physics-Based Model Links Data to Transport Parameters

Forward Model

$$\frac{dx(t)}{dt} = \mathbf{g}(x(t), u(t), \theta)$$

$$\hat{\mathbf{y}}(t) = \mathbf{f}(x(t), \hat{\theta})$$

$$\mathbf{y}(t) = \hat{\mathbf{y}}(t) + \epsilon(t)$$

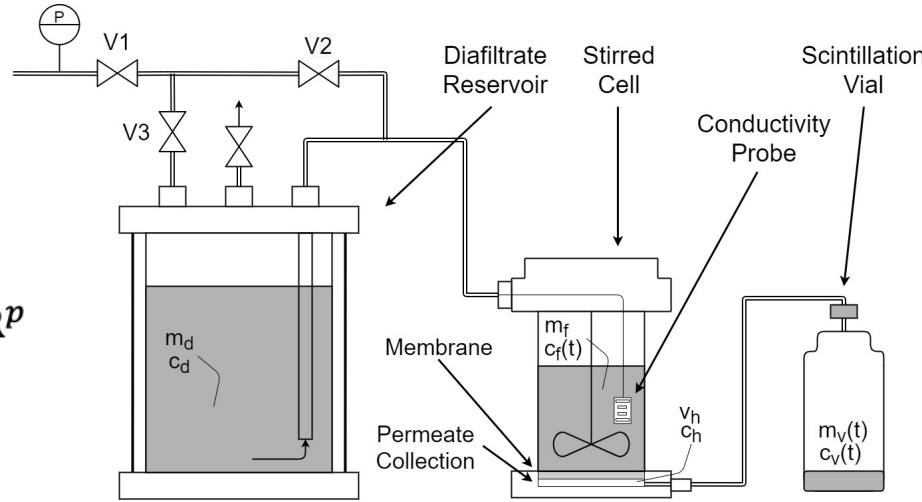
$$\mathbf{x}(t) \in \mathbb{R}^n; \quad \mathbf{u}(t) \in \mathbb{R}^m; \quad \theta \in \mathbb{R}^p$$

$$\epsilon(t) \sim \mathcal{N}(0, s)$$

$$\theta = \begin{bmatrix} L_p \\ B \\ \sigma \end{bmatrix}; \quad \mathbf{y}(t) = \begin{bmatrix} m_v \\ c_v \\ c_f \end{bmatrix}$$

ΔP : applied pressure drop
 n : number of dissolved species
 R : gas constant
 T : temperature
 A_m : area of the membrane

k : Mass transfer coefficient of interest
 b : Diameter of the stirred cell
 D : Diffusion coefficient of solute in solvent
 ν : Kinematic viscosity of the solvent
 ν^0 : Average velocity within the system



Water flux across the membrane

$$J_w = L_p(\Delta P - \sigma \Delta \pi)$$

$$\Delta \pi = nRT(c_{in} - c_h)$$

Solute flux across the membrane

$$J_s = B(c_{in} - c_h)$$

Concentration polarization

$$\frac{c_{in} - c_h}{c_f - c_h} = \exp\left(\frac{J_w}{k}\right)$$

$$\frac{k_b}{D} = \left(\frac{b\nu^0}{\nu}\right)^{0.57} \left(\frac{\nu}{D}\right)^{0.33}$$

Mass balance in dialysate reservoir

$$\frac{dm_d}{dt} = -A_m \rho J_w$$

Constant concentration in dialysate reservoir

$$\frac{dc_d}{dt} = 0$$

Constant mass in feed cell

$$\frac{dm_f}{dt} = 0$$

Solute balance in feed cell

$$\frac{d(c_f m_f)}{dt} = A_m \rho (J_w c_d - J_s)$$

Constant mass in holdup

$$\frac{dm_h}{dt} = 0$$

Solute balance in holdup

$$\frac{d(c_h m_h)}{dt} = A_m \rho J_s + \frac{dm_d}{dt} c_h$$

Mass balance in collecting vial

$$\frac{dm_v}{dt} = -\frac{dm_d}{dt} = A_m \rho J_w$$

Solute balance in collecting vial

$$\frac{d(c_v m_v)}{dt} = -\frac{dm_d}{dt} c_h = A_m \rho J_w c_h$$

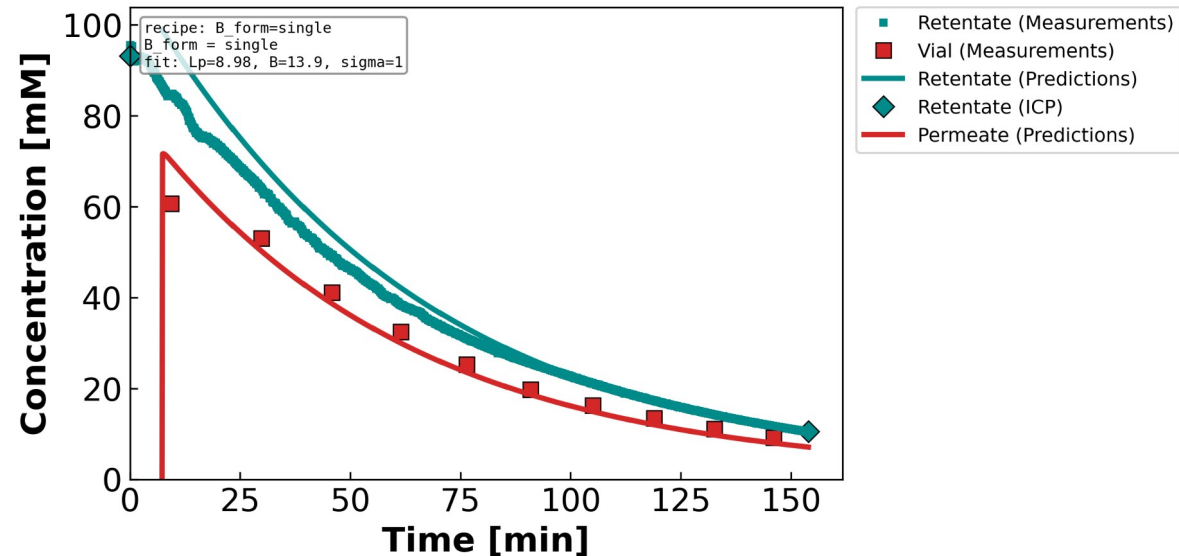
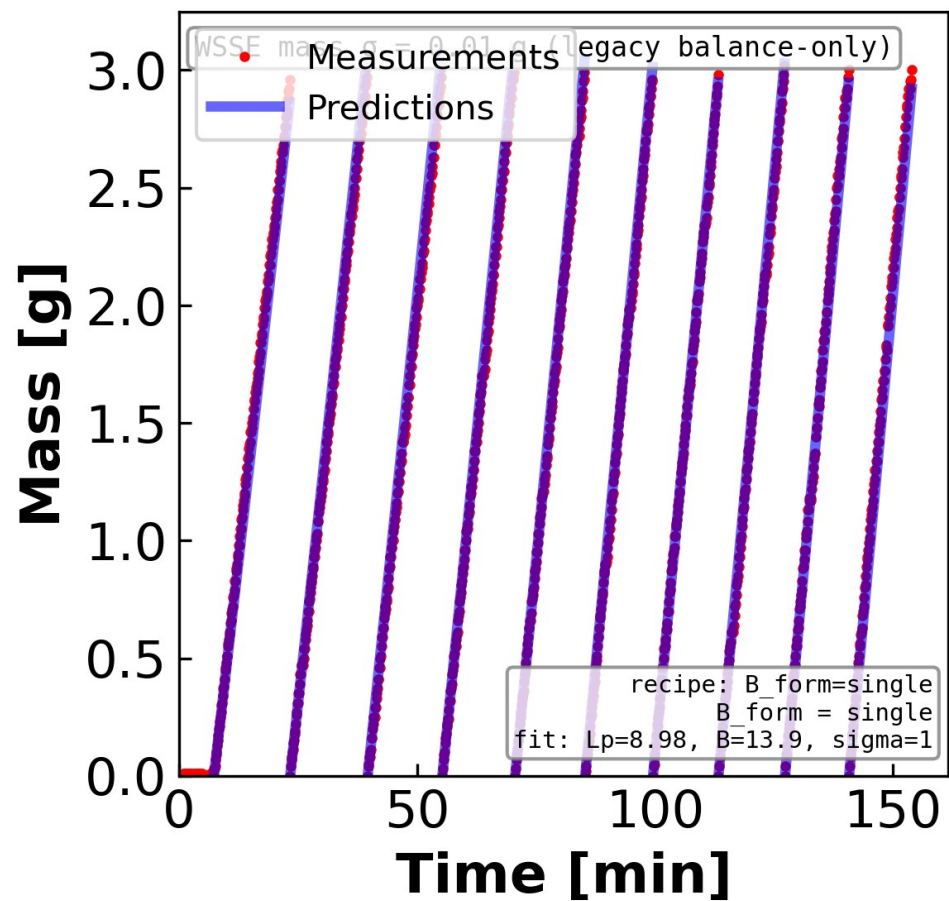


Figure 1. Measured and model-predicted time profiles for a representative diluting NaCl diafiltration experiment. (Left) total retentate mass and (right) solute concentration in the retentate and permeate streams versus time; symbols are measurements — the retentate inductively-coupled-plasma (ICP) reference points and the time-resolved permeate vial samples — and the curves are the lumped-parameter model evaluated at the fitted parameters reported in the inset ($L_p = 8.98 \text{ L m}^{-2} \text{ h}^{-1} \text{ bar}^{-1}$, $B = 13.88 \text{ } \mu\text{m s}^{-1}$, $\sigma = 1$). A single concentration-independent solute permeability reproduces the mass decline and both concentration histories, with the retentate and permeate plotted in distinct colors; the reflection coefficient is forced to its upper bound ($\sigma = 1$).

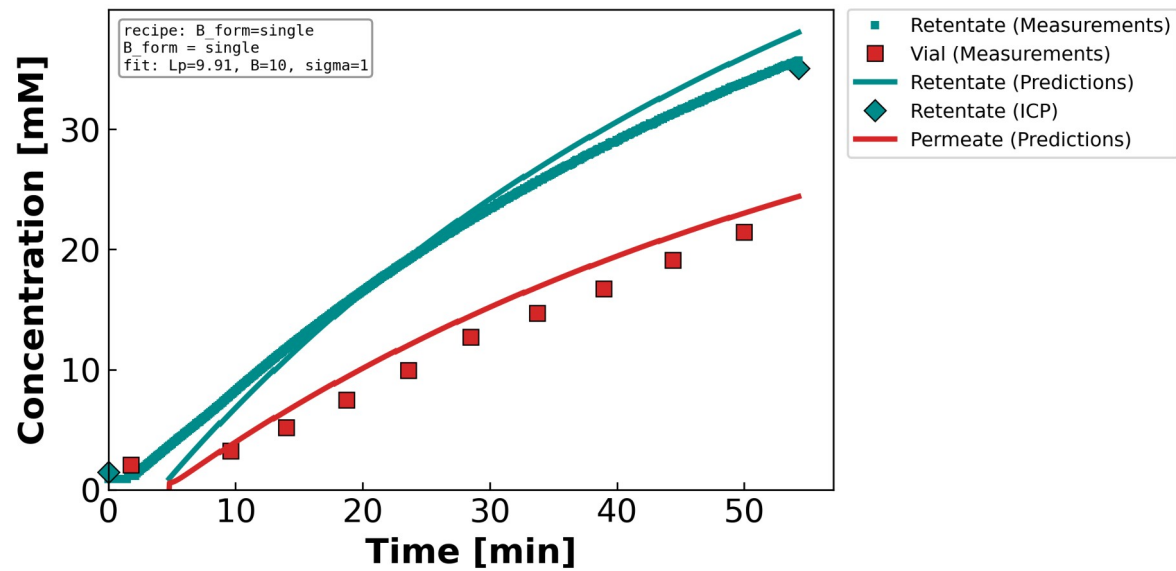
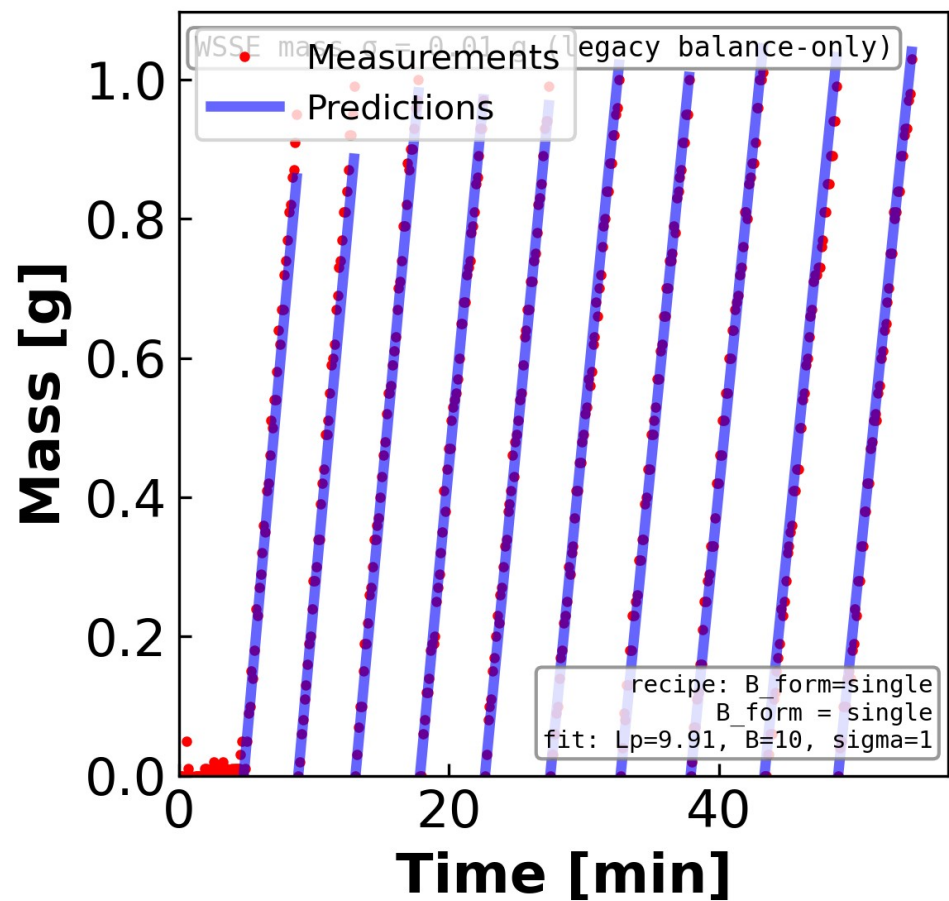


Figure 2. Measured and model-predicted time profiles for a representative concentrating NaCl diafiltration experiment. (Left) total retentate mass and (right) solute concentration in the retentate and permeate streams versus time; symbols are measurements — the retentate inductively-coupled-plasma (ICP) reference points and the time-resolved permeate vial samples — and the curves are the lumped-parameter model evaluated at the fitted parameters reported in the inset ($L_p = 9.91 \text{ L m}^{-2} \text{ h}^{-1} \text{ bar}^{-1}$, $B = 10.00 \text{ } \mu\text{m s}^{-1}$, $\sigma = 1$). A single concentration-independent solute permeability reproduces the mass decline and both concentration histories, with the retentate and permeate plotted in distinct colors; the reflection coefficient is forced to its upper bound ($\sigma = 1$).

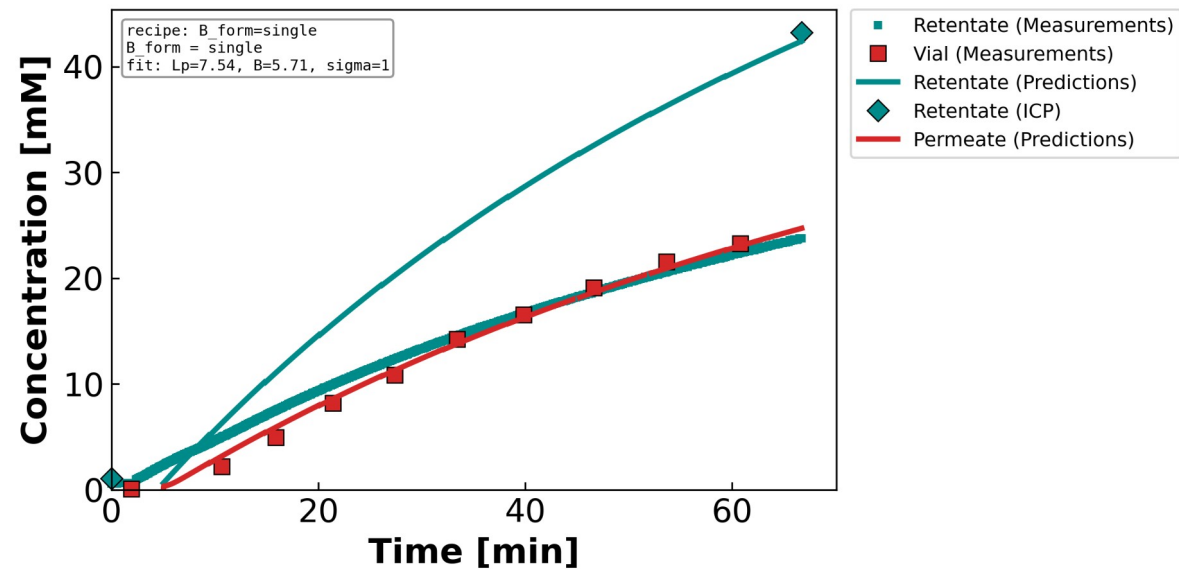
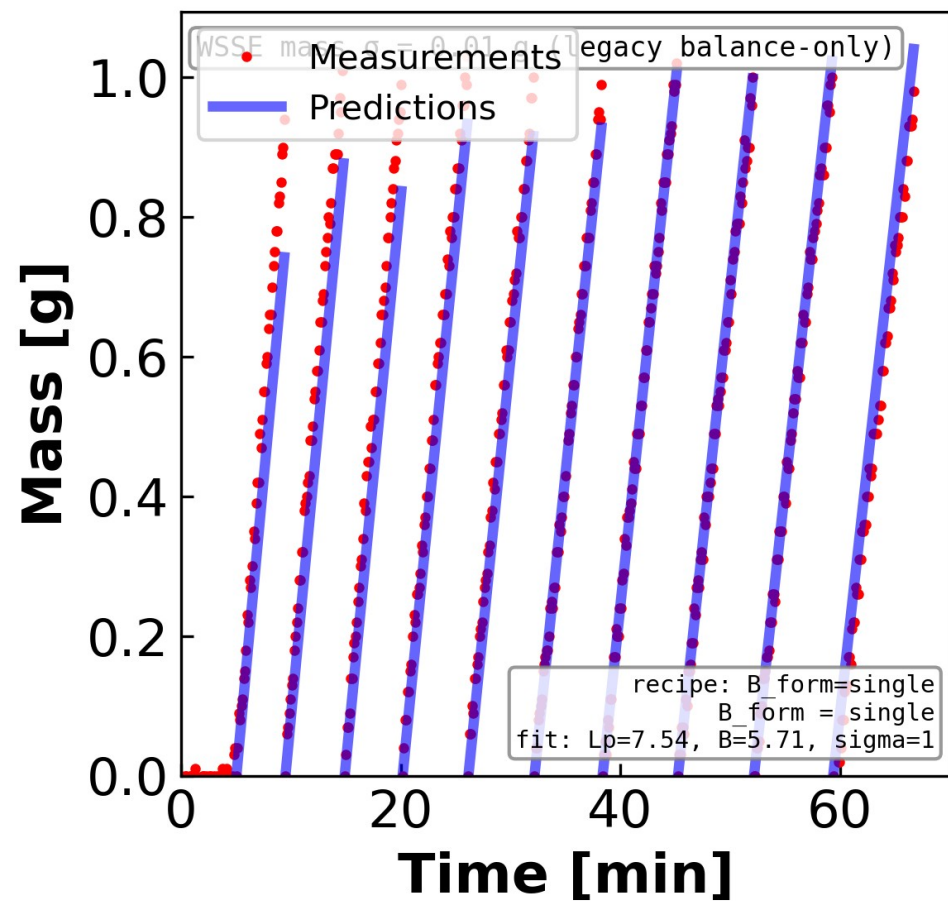


Figure 3. Measured and model-predicted time profiles for a representative concentrating CaCl_2 diafiltration experiment. (Left) total retentate mass and (right) solute concentration in the retentate and permeate streams versus time; symbols are measurements — the retentate inductively-coupled-plasma (ICP) reference points and the time-resolved permeate vial samples — and the curves are the lumped-parameter model evaluated at the fitted parameters reported in the inset ($L_p = 7.54 \text{ L m}^{-2} \text{ h}^{-1} \text{ bar}^{-1}$, $B = 5.71 \text{ } \mu\text{m s}^{-1}$, $\sigma = 1$). A single concentration-independent solute permeability reproduces the mass decline and both concentration histories, with the retentate and permeate plotted in distinct colors; the reflection coefficient is forced to its upper bound ($\sigma = 1$).

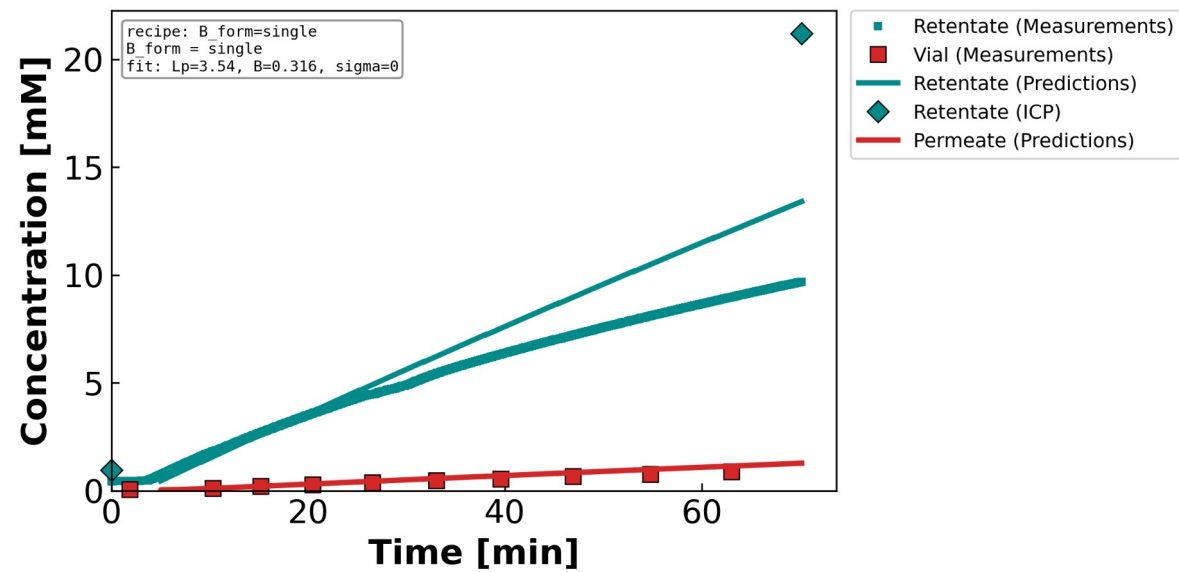
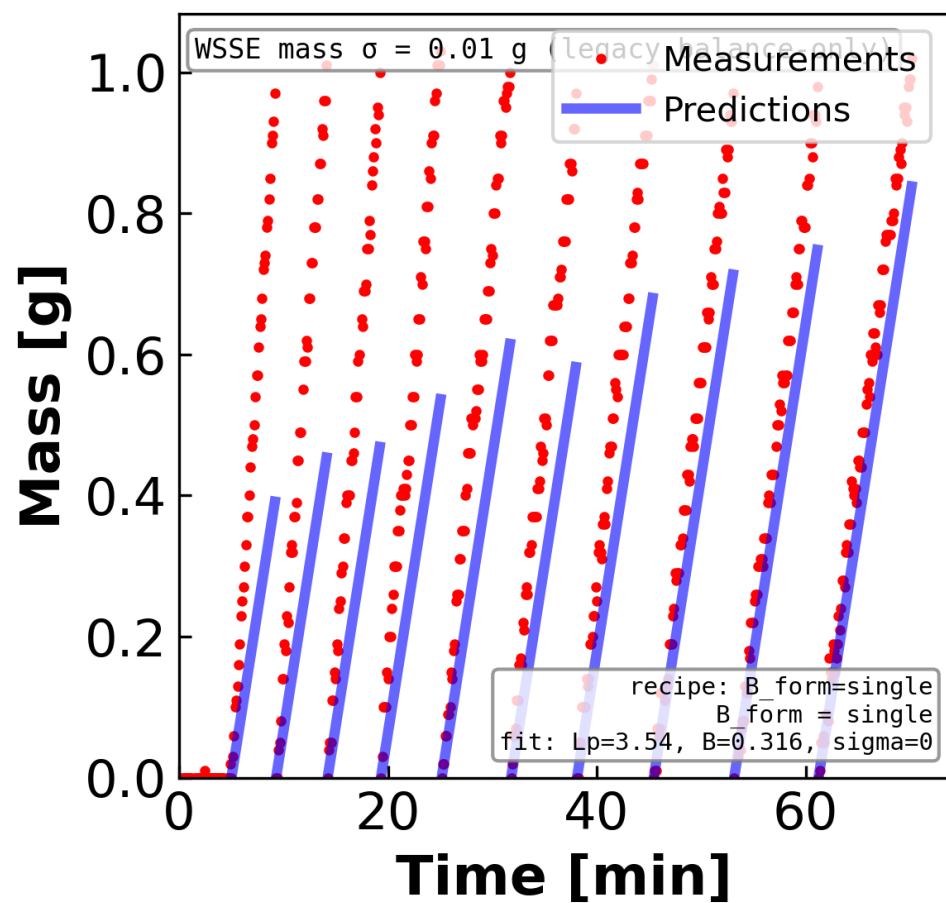


Figure 4. Measured and model-predicted time profiles for a representative concentrating LaCl_3 diafiltration experiment. (Left) total retentate mass and (right) solute concentration in the retentate and permeate streams versus time; symbols are measurements — the retentate inductively-coupled-plasma (ICP) reference points and the time-resolved permeate vial samples — and the curves are the lumped-parameter model evaluated at the fitted parameters reported in the inset ($L_p = 3.54 \text{ L m}^{-2} \text{ h}^{-1} \text{ bar}^{-1}$, $B = 0.32 \mu\text{m s}^{-1}$, $\sigma = 0$). A single concentration-independent solute permeability reproduces the mass decline and both concentration histories, with the retentate and permeate plotted in distinct colors; the reflection coefficient is forced to its lower bound ($\sigma = 0$). This trivalent-salt experiment is the most demanding case in the campaign and shows the largest residuals of the four.

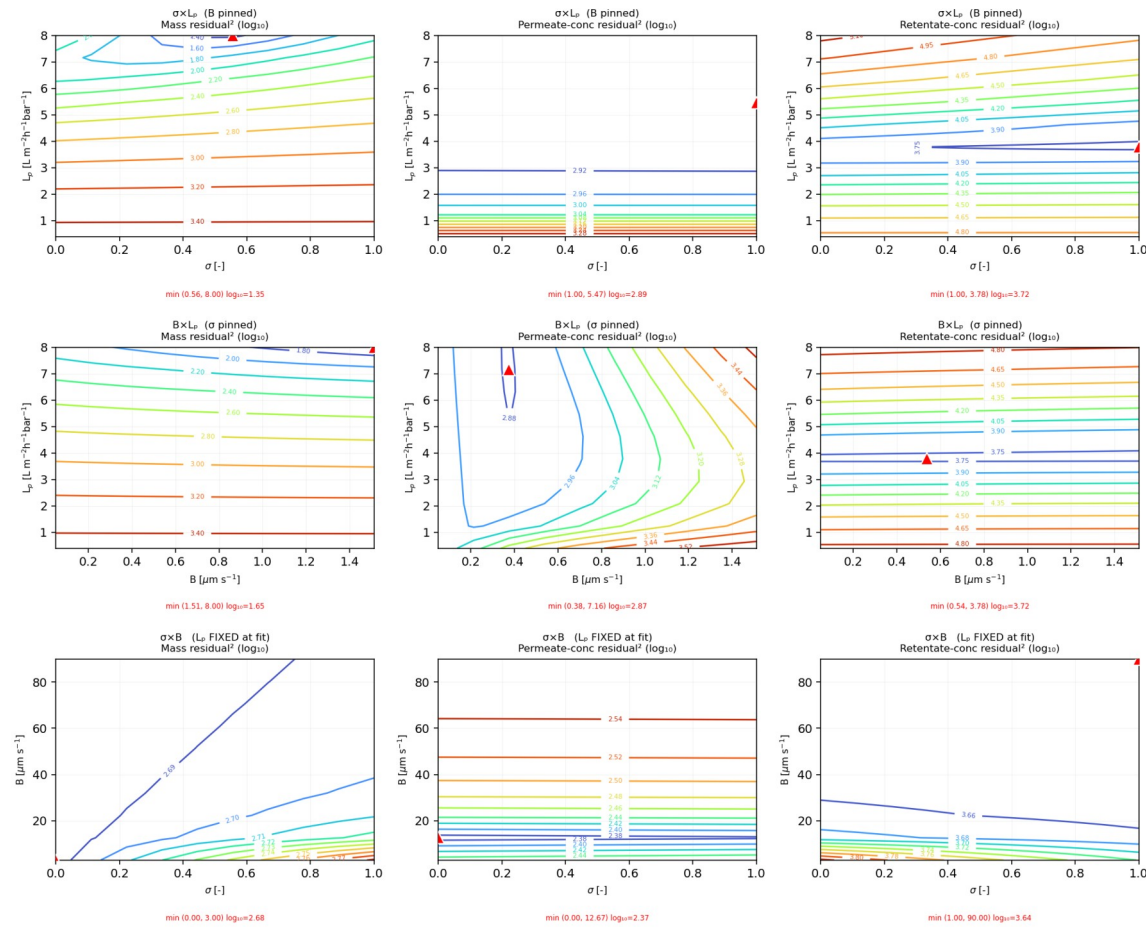


Figure 5. Log-transformed residual-squared contours for a representative concentrating CaCl₂ diafiltration experiment, generated by a 2-D grid search in which model predictions are computed for all combinations of two lumped transport parameters while the third is held at its fitted value. Columns correspond to the mass, permeate-concentration, and retentate-concentration objectives; rows correspond to the $\sigma \times L_p$, $B \times L_p$, and $\sigma \times B$ planes. The red triangle marks the minimum of each surface. The horizontal iso-residual bands in the upper two rows localize a unique hydraulic permeability L_p that is insensitive to σ and B , whereas the $\sigma \times B$ plane (bottom row) displays an extended valley, indicating that the reflection coefficient and solute permeability trade off and are not jointly identified from a single experiment. This well-determined L_p is what licenses fixing it in the $\sigma \times B$ maps that follow.

diluting NaCl (MC3.07.22.24) — $\sigma \times B$ identifiability, hydraulic permeability fixed at its identified optimum $L_p = 8.98 \text{ L m}^{-2} \text{ h}^{-1} \text{ bar}^{-1}$

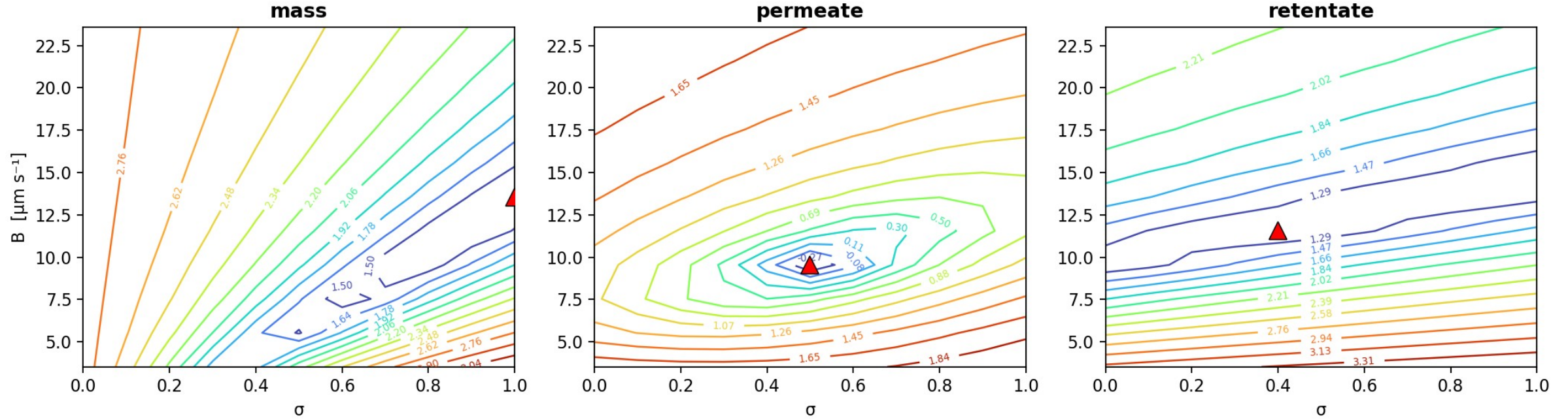


Figure 6. Identifiability of the reflection coefficient σ and the solute permeability B for a representative diluting NaCl experiment, with the hydraulic permeability fixed at its identified optimum $L_p = 8.98 \text{ L m}^{-2} \text{ h}^{-1} \text{ bar}^{-1}$ (well determined and insensitive to σ and B ; Figure 5). The three panels are the base-10 logarithm of the weighted sum-of-squared-error (WSSE) surface for each measured response — labelled mass, permeate, and retentate — computed by forward simulation over the $\sigma \times B$ grid at that fixed L_p ; the red triangle marks each surface's minimum. Each surface constrains B through a narrow valley but is only weakly sensitive to σ , so the three responses reach their minima at different σ values along the boundaries; the reflection coefficient is therefore not jointly identified, and the single-experiment optimum places σ at its upper bound ($\sigma = 1$). The solute permeability is well determined while σ is not recoverable from one experiment.

concentrating NaCl (MC2.05.07.24) — $\sigma \times B$ identifiability, hydraulic permeability fixed at its identified optimum $L_p = 9.91 \text{ L m}^{-2} \text{ h}^{-1} \text{ bar}^{-1}$

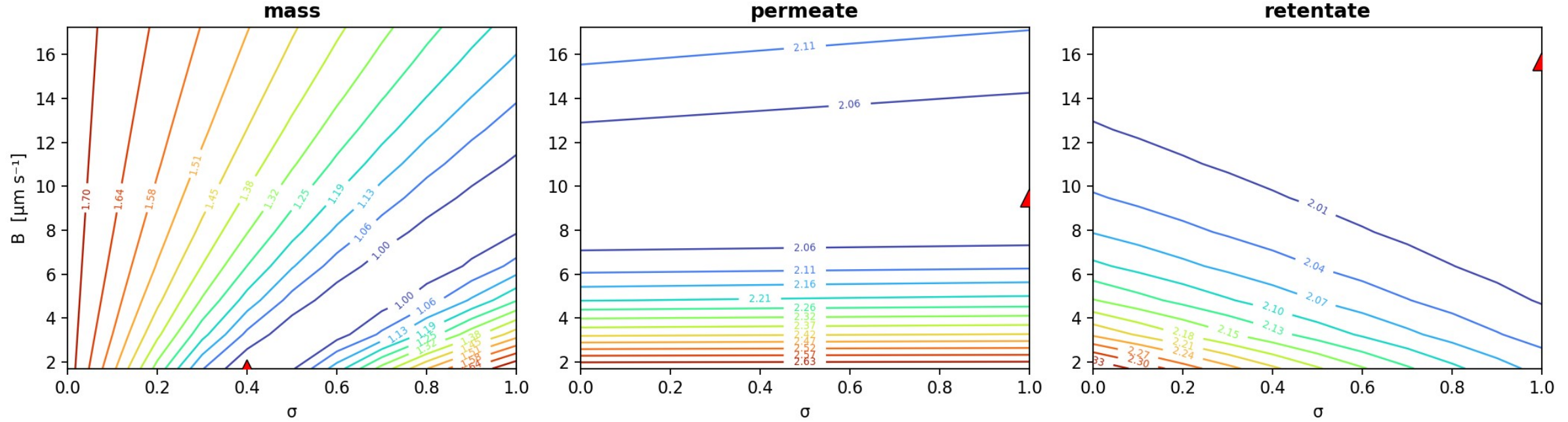


Figure 7. Identifiability of the reflection coefficient σ and the solute permeability B for a representative concentrating NaCl experiment, with the hydraulic permeability fixed at its identified optimum $L_p = 9.91 \text{ L m}^{-2} \text{ h}^{-1} \text{ bar}^{-1}$ (well determined and insensitive to σ and B ; Figure 5). The three panels are the base-10 logarithm of the weighted sum-of-squared-error (WSSE) surface for each measured response — labelled mass, permeate, and retentate — computed by forward simulation over the $\sigma \times B$ grid at that fixed L_p ; the red triangle marks each surface's minimum. Each surface constrains B through a narrow valley but is only weakly sensitive to σ , so the three responses reach their minima at different σ values along the boundaries; the reflection coefficient is therefore not jointly identified, and the single-experiment optimum places σ at its upper bound ($\sigma = 1$). The solute permeability is well determined while σ is not recoverable from one experiment.

concentrating CaCl_2 (MC2.05.07.24) — $\sigma \times B$ identifiability, hydraulic permeability fixed at its identified optimum $L_p = 7.54 \text{ L m}^{-2} \text{ h}^{-1} \text{ bar}^{-1}$

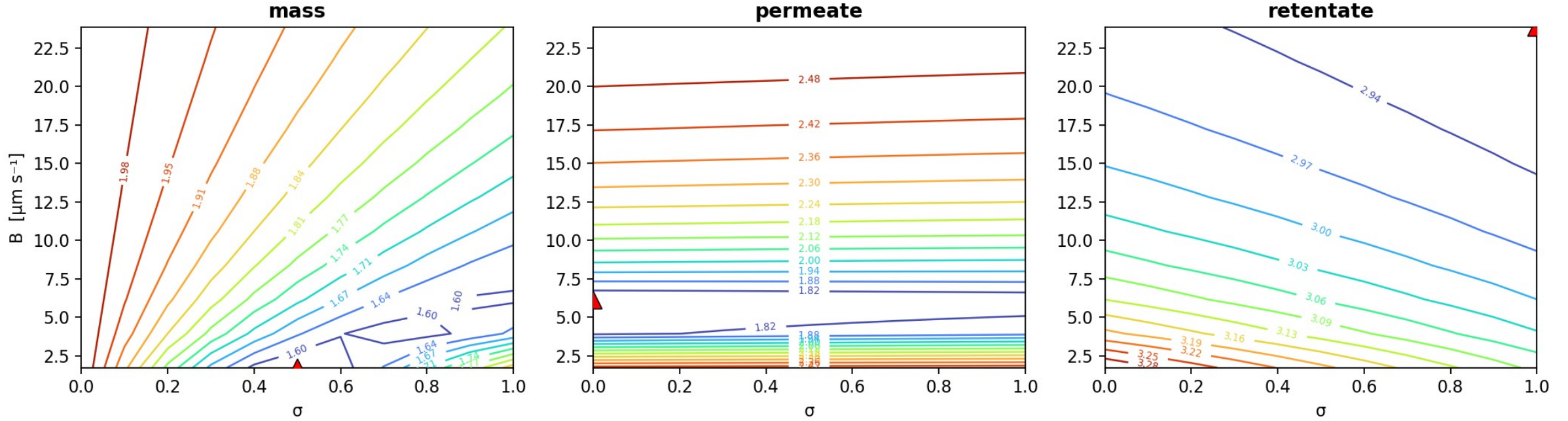


Figure 8. Identifiability of the reflection coefficient σ and the solute permeability B for a representative concentrating CaCl_2 experiment, with the hydraulic permeability fixed at its identified optimum $L_p = 7.54 \text{ L m}^{-2} \text{ h}^{-1} \text{ bar}^{-1}$ (well determined and insensitive to σ and B ; Figure 5). The three panels are the base-10 logarithm of the weighted sum-of-squared-error (WSSE) surface for each measured response — labelled mass, permeate, and retentate — computed by forward simulation over the $\sigma \times B$ grid at that fixed L_p ; the red triangle marks each surface's minimum. Each surface constrains B through a narrow valley but is only weakly sensitive to σ , so the three responses reach their minima at different σ values along the boundaries; the reflection coefficient is therefore not jointly identified, and the single-experiment optimum places σ at its upper bound ($\sigma = 1$). The solute permeability is well determined while σ is not recoverable from one experiment.

concentrating LaCl_3 (MC2.05.21.24) — $\sigma \times B$ identifiability, hydraulic permeability fixed at its identified optimum $L_p = 3.54 \text{ L m}^{-2} \text{ h}^{-1} \text{ bar}^{-1}$

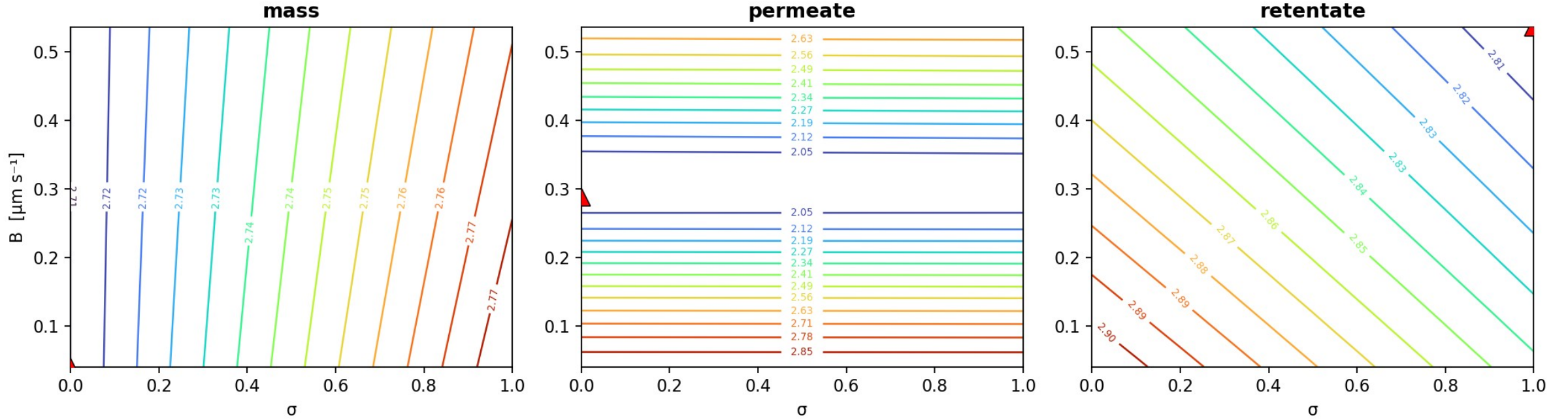


Figure 9. Identifiability of the reflection coefficient σ and the solute permeability B for a representative concentrating LaCl_3 experiment, with the hydraulic permeability fixed at its identified optimum $L_p = 3.54 \text{ L m}^{-2} \text{ h}^{-1} \text{ bar}^{-1}$ (well determined and insensitive to σ and B ; Figure 5). The three panels are the base-10 logarithm of the weighted sum-of-squared-error (WSSE) surface for each measured response — labelled mass, permeate, and retentate — computed by forward simulation over the $\sigma \times B$ grid at that fixed L_p ; the red triangle marks each surface's minimum. Each surface constrains B through a narrow valley but is only weakly sensitive to σ , so the three responses reach their minima at different σ values along the boundaries; the reflection coefficient is therefore not jointly identified, and the single-experiment optimum places σ at its lower bound ($\sigma = 0$). The solute permeability is well determined while σ is not recoverable from one experiment.